SEHAT-SETU: DP-900 Data Architecture

Project-Based Learning (PBL) Report | Core Azure Data Fundamentals

A foundational requirement of the SEHAT-SETU platform is the efficient and secure processing of disparate healthcare data types. In alignment with Microsoft Azure Data Fundamentals (DP-900), the system implements a polyglot persistence strategy, deliberately separating structured relational clinical data from unstructured administrative and audit media to optimize both compute performance and storage costs.

1. Structured Data Management

Definition: Structured data adheres to a rigid, predefined schema (rows and columns) and is managed via Relational Database Management Systems (RDBMS). It is highly organized and easily queried using Structured Query Language (SQL).

Within SEHAT-SETU, all core transactional and relationship data is classified as structured. To manage this, the platform utilizes Azure SQL Database.

Data Entities	Administrative Units (districts), Healthcare Institutions (facilities), Master Specialties (services), User Profiles & RBAC Credentials (users), Capacity Declarations (facility_capacity), Patient Referral Transfers (referrals), and System Activity Logs (activity_logs).
Schema & Integrity	The database strictly enforces a Third Normal Form (3NF) architecture across 7 normalized tables. Relationships, such as a facility staff member creating an outbound patient referral or logging specialist shortages, are governed by explicit Foreign Key constraints, check constraints, and unique keys, preventing orphaned records and maintaining referential integrity.
Operational Benefits	• **ACID Compliance:** Ensures that multi-step database transactions (e.g., dispatching an urgent patient transfer, logging the referral status, and recording audit telemetry in activity_logs) are processed reliably and atomically. • **Complex Querying:** Enables deep analytical queries, such as executing multi-table INNER JOIN operations between referrals, facilities, and services grouped by district_id to correlate outbound referral surges directly against certified specialist vacancies instantly.

2. Unstructured Data Management

Definition: Unstructured data lacks a predefined data model or schema. It encompasses raw binary files such as images, videos, or documents, which cannot be efficiently stored or queried in traditional relational tables.

SEHAT-SETU is a healthcare referral intelligence and gap identification platform reliant on qualitative administrative documentation, including quarterly facility gap audit reports, clinical transfer guidelines, and equipment maintenance circulars. Storing these binary and long-form text files (PDFs/DOCX/TXT) within Azure SQL would result in severe database bloat, increased latency, and exorbitant compute costs. Therefore, this data is classified as unstructured and routed to Azure Blob Storage.

Data Entities	Quarterly Chief Medical Officer (CMO) Facility Gap Audit Memorandums, Inter-Facility Transfer Protocol SOPs, Clinical Inspection Evaluations, and Diagnostic Equipment Downtime Circulars.
Storage Mechanism	Documents and audit files are uploaded directly as binary objects (blobs) into designated Azure storage containers (sehatsetu-blobs/unstructured/). The data possesses no relational schema; rather, it is identified by unique object URIs.
Operational Benefits	• **Infinite Scalability:** Blob storage horizontally scales to accommodate terabytes of historical audit documentation without requiring database provisioning or indexing. • **Cost Efficiency:** Object storage is significantly cheaper per gigabyte compared to premium relational database storage (utilizing Azure Hot, Cool, and Archive tiers for document lifecycle management). • **API Interoperability:** Azure Blob Storage generates direct SAS URLs for each audit dossier, allowing the SEHAT-SETU backend to pass secure document links directly to district healthcare officers for instant review without loading large files into backend server memory.

3. Architectural Synergy

By cleanly separating these data domains, SEHAT-SETU achieves optimal cloud efficiency. The Azure SQL Database remains lightweight and highly responsive for user interactions, authentication routing, and structured referral analytics, while Azure Blob Storage acts as a highly durable, low-cost repository serving the heavy administrative audit files and clinical policy guidelines. This dual-pipeline approach embodies the core architectural principles taught in the DP-900 curriculum.